

PRODUCT CASE STUDY

How We Reduced Checkout Drop-Off by 28%

A data-driven product case study on optimising the mobile checkout funnel for an e-commerce marketplace

Role	Product Management & Analysis
Domain	E-commerce / Marketplace
Timeline	3 months approx (Discovery to Launch)
Team	PM, 2 Engineers, 1 Designer, 1 Data Analyst
Outcome	28% reduction in checkout drop-off; +₹4.2Cr GMV/month

28% Checkout Conversion Lift 46% → 59% cart-to-purchase	-60% Payment Page Load Time 4.8s → 1.9s (P75)	₹4.2Cr Monthly GMV Uplift ₹15Cr → ₹19.2Cr / month
--	--	--

1. Context & Background

A mid-sized e-commerce marketplace (fashion + lifestyle, 2.4M MAUs) was experiencing a persistent and worsening checkout abandonment problem on mobile. Despite healthy top-of-funnel traffic and strong add-to-cart rates, a significant portion of users were dropping off between cart and payment confirmation — representing tens of crores in lost monthly GMV.

As PM on this initiative, I was tasked with identifying root causes, designing and prioritising solutions, and coordinating a cross-functional sprint to ship improvements within a single quarter. The mandate was clear: move the needle on mobile conversion without a full redesign.



Core Question: Why are users adding items to cart but abandoning before completing payment — and what are the highest-leverage fixes we can ship in one sprint?

2. Problem Discovery

2.1 Funnel Analysis

Starting with funnel analysis across 4 weeks of session data (n = 340,000 checkout initiations), I mapped where users were dropping off at each step. The payment method selection step was the critical leak — mobile users were abandoning here at a rate 3x worse than desktop.

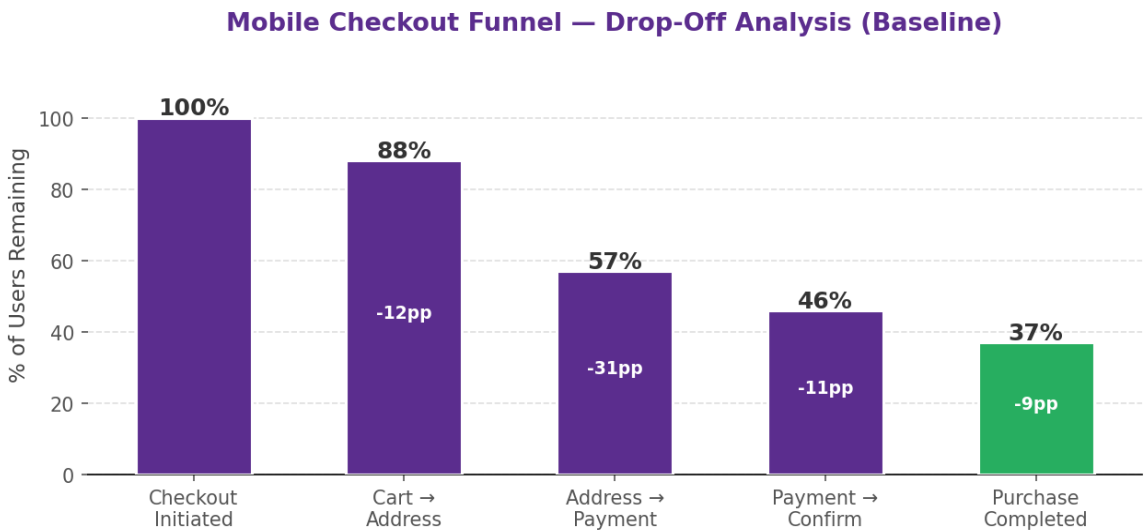


Fig. 1 — Mobile Checkout Funnel: Drop-off at Each Stage (Baseline). Numbers in bars show point-to-point loss.

Funnel Step	Drop-off Rate	Volume Lost/Week	vs. Desktop
Cart → Address entry	12%	~6,800 sessions	2x worse on mobile
Address → Payment method	31%	~18,200 sessions	3.1x worse on mobile
Payment → Confirm	19%	~8,900 sessions	1.4x worse on mobile
Overall cart-to-purchase	54% abandoned	~31,800 sessions	Industry avg: 38%

2.2 Qualitative Research

I ran 14 user interviews (8 mobile, 6 desktop) and reviewed 200 session recordings. Four root-cause themes emerged clearly and consistently across user segments:

"I had my card ready but the payment page kept loading. I just gave up and went back to browsing." — Mobile user, 26F

"There were so many payment options I didn't know which to trust. I just closed the app." — Mobile user, 34M

Root Cause	Finding	Severity
------------	---------	----------

Performance	Payment page P75 load time was 4.8s — above 3s drop-off threshold	Critical
Choice overload	17 payment options shown with no hierarchy — overwhelmed first-time users	High
Missing trust signals	No security badges or 'safe checkout' copy near CTA on mobile	High
Saved card UX	Returning users couldn't see saved cards without scrolling past all options	Medium

3. Prioritisation

I mapped identified issues against an Impact × Effort matrix and used the RICE framework to sequence solutions for the sprint. The goal was to identify the highest-leverage changes we could ship safely within 8 weeks.

Opportunity	Reach	Impact	Confidence	Effort	RICE Score	Priority
Lazy-load payment options	High	High	90%	S (3d)	8.1	P0
Personalised payment defaults	High	High	80%	M (1w)	7.2	P0
Reduce to top 5 + 'more'	High	Medium	85%	S (2d)	6.8	P0
Add trust badges + copy	High	Medium	70%	XS (1d)	6.1	P1
One-tap saved card	Medium	High	75%	M (1w)	5.6	P1
Guest checkout shortcut	Low	Medium	65%	L (2w)	2.8	P2

4. Solution Design & What We Built

4.1 Sprint 1 — P0 Changes

- **Surface the user's most-used payment method pre-selected.** For new users, default to UPI (highest completion rate in demographic data). Reduced decision fatigue at the highest drop-off point.**Smart Payment Defaults:**
- **Show top 5 payment options (UPI, saved cards, COD, net banking, EMI) with a collapsible 'More options'.** Removed visual noise for 80% of users while preserving full flexibility.**Progressive Disclosure:**

- **Lazy-load below-the-fold payment options. Pre-fetch gateway assets on add-to-cart (not on checkout initiation). Reduced P75 load time from 4.8s to 1.9s.** Performance Fix:

4.2 Sprint 2 — P1 Changes

- Trust signals: Added PCI-DSS badge + '100% Secure Payments' microcopy above the payment CTA
- One-tap saved card: Returning users with a saved card see it surfaced above all options



Design Principle: Every change was validated on a prototype with 5 users before engineering build. We killed 2 candidate ideas at prototype stage — saving approximately 3 weeks of wasted build time.

5. Results

The combination of P0 and P1 changes produced a compounding improvement across all key checkout metrics. The payment step — our highest-priority drop-off point — improved by 39%, and overall mobile GMV grew by ₹4.2Cr per month within 8 weeks of the final rollout.

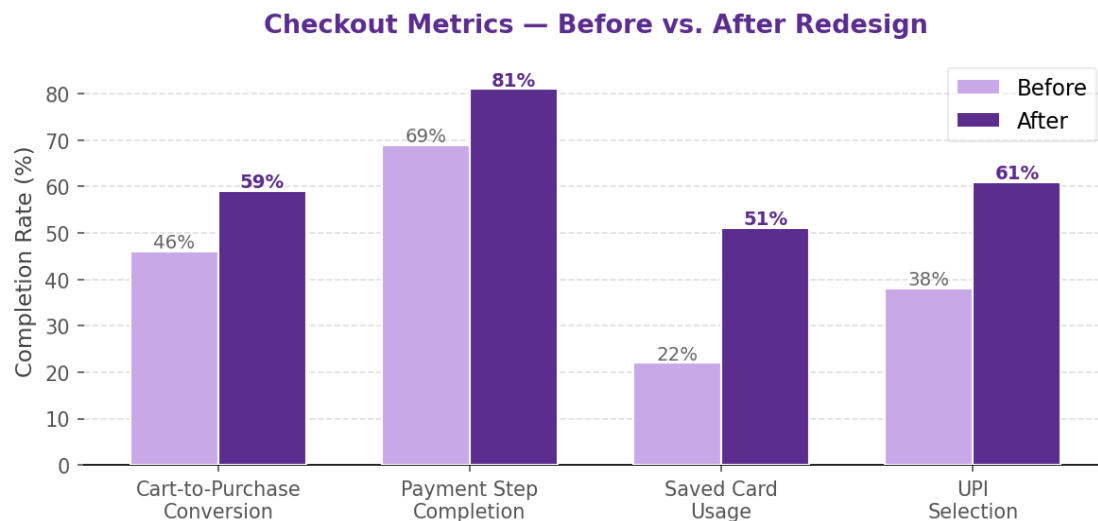


Fig. 2 — Key Checkout Metrics: Before vs. After Redesign

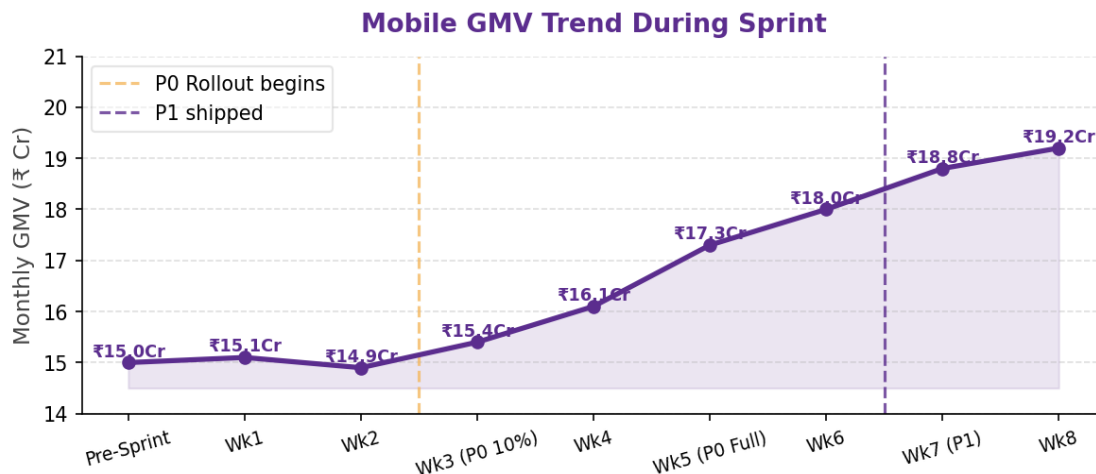


Fig. 3 — Mobile GMV Trend During Sprint (Week-by-Week). P0 full rollout at Week 5 produced the sharpest inflection.

Metric	Before	After	Change
Cart-to-purchase (mobile)	46%	59%	+28% relative improvement
Payment page P75 load time	4.8 seconds	1.9 seconds	−60% latency reduction
Monthly GMV (mobile)	₹15Cr	₹19.2Cr	+₹4.2Cr / month
Payment step abandonment	31%	19%	−39% drop in abandonment
UPI selection rate	38%	61%	+23pp (smart default effect)

6. Key Learnings

6.1 What Worked

- Starting with data to locate the exact funnel leak saved weeks of guesswork — we knew the payment step was the problem before writing a single line of code
- Progressive disclosure was the single highest-RICE change — simplicity wins at scale, always
- Pre-fetching assets on add-to-cart (not checkout) was a counter-intuitive but high-impact engineering idea that only surfaced because engineers were in the discovery process

6.2 What I'd Do Differently

- Run the A/B test for 2 extra weeks to capture full weekend shopping cycles — our initial read was slightly overstated due to weekday-skewed timing
- Involve the trust & safety team earlier — the security badge copy needed legal review, which delayed P1 by 4 days
- Set up a post-launch monitoring dashboard before shipping — we were slower to catch an edge-case bug on iOS 15

6.3 The Broader PM Lesson

This case study is ultimately about the value of structured problem decomposition. The 28% improvement was not the result of a single brilliant idea — it was the compounding effect of six well-reasoned, well-sequenced changes, each validated before scaling. The PM's job was to keep the team focused on the right problems in the right order, and to resist the temptation to redesign everything at once.



Takeaway: Good product thinking is not about having the best ideas. It is about asking the right questions, letting data lead, and making ruthless prioritisation decisions with incomplete information.